1 A student is investigating the angle at which a glass cylinder containing oil topples, as shown in Fig. 1.1.

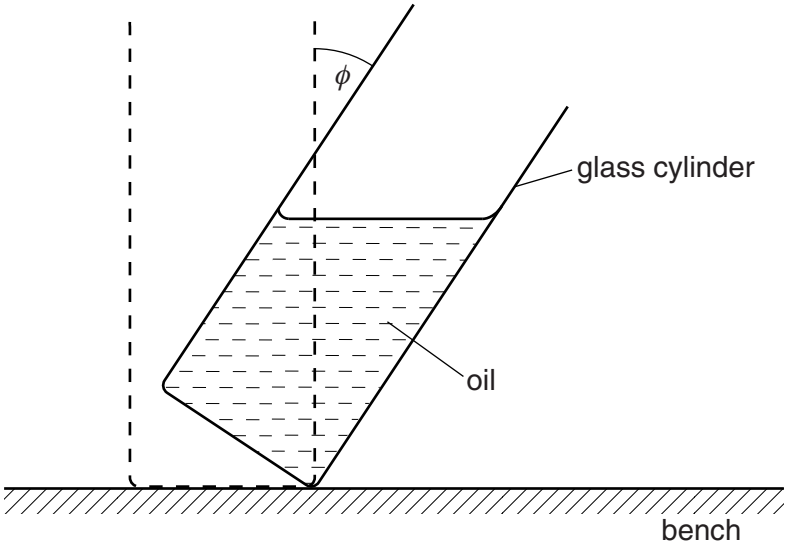

Fig. 1.1

A cylinder containing a mass m of oil can be tilted through a maximum angle ϕ from the vertical before it topples.

It is suggested that the relationship between m and ϕ is

$$\frac{1}{\tan \phi} = \frac{am}{\rho d^3} + b$$

where d is the diameter of the cylinder, ρ is the density of the oil and a and b are constants.

Design a laboratory experiment to test the relationship between ϕ and m. Explain how your results could be used to determine values for a and b. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

[15]

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